

9. (a) Ethanedioic acid (oxalic acid) has the formula $(\text{COOH})_2$.

It can be made in the laboratory by oxidising sugar with a powerful oxidising agent (shown as [O] in the equation below).

- (i) Balance the equation. [1]



- (ii) After purification, the oxalic acid is produced as white crystals of a hydrate, $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. On heating this hydrate, water is lost, leaving anhydrous oxalic acid.



In an experiment 5.67 g of the hydrate were heated to constant mass, giving 4.05 g of the anhydrous acid.

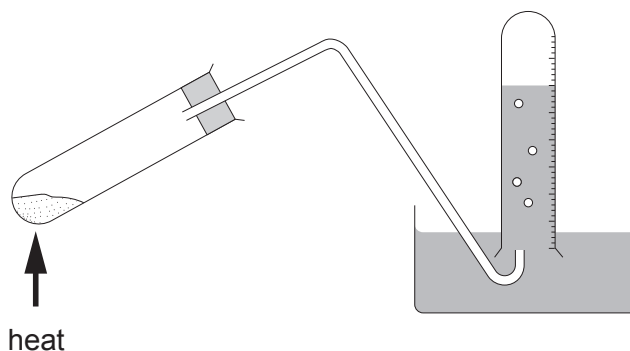
Calculate the value of x in $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. You **must** show your working. [3]

$x = \dots\dots\dots$

- (b) Another method of producing oxalic acid in the laboratory is to heat potassium methanoate, HCOOK . This produces potassium oxalate, which is then acidified to give the acid.



Some students carried out this experiment using the apparatus below. They collected the hydrogen gas produced over water.



A few students obtained a smaller volume of hydrogen than expected.

Suggest **two** practical reasons why this may have occurred, apart from leaks. [2]

1

2

- (c) Gareth was asked to analyse a mixture of potassium methanoate and potassium oxalate. He weighed out 4.69 g of the mixture and carried out the method below.

- The mixture was completely dissolved in distilled water.
- Calcium chloride solution was added to this solution (only calcium oxalate was precipitated).
- The mixture was filtered and the precipitated calcium oxalate washed and dried.

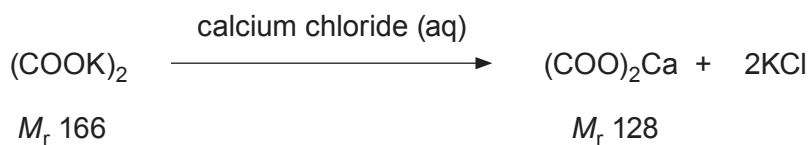
- (i) Suggest how Gareth would know when enough calcium chloride solution had been added to react with all of the potassium oxalate present. [2]

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- (ii) In his experiment Gareth obtained 2.49 g of pure dry calcium oxalate (M_r 128).



Calculate the percentage of potassium oxalate in the mixture, giving your answer to the appropriate number of significant figures. [3]

Percentage of potassium oxalate = %

